



TRINITY GRAMMAR SCHOOL

Mathematics Department

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(NESA Student Number)

2025

Year 11

Mathematics Advanced

YEARLY EXAMINATION

ASSESSMENT TASK 3

Date of Assessment Task: Friday, 29 August 2025

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Erasable pens and all electronic devices (or timekeepers), unless otherwise advised, are not permitted in this Assessment
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- *Write your NESA Student Number at the top of this page*

Total marks:
70

Section I – 10 marks (pages 2 – 5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 6 – 20)

- Attempt Questions 11 – 31
- Allow about 1 hour and 45 minutes for this section

- **Year 11 Assessment Weighting:** 60%

Section I

10 marks

Attempt Questions 1–10

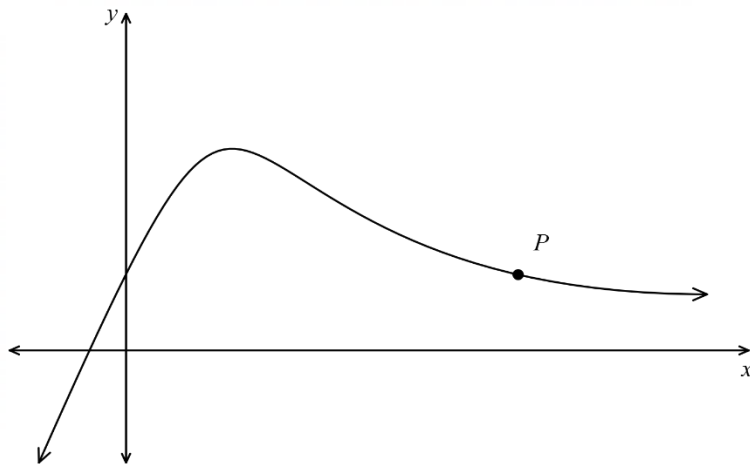
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 What are the solutions to the quadratic equation $3x^2 - x - 4 = 0$?

- A. $x = -1$ and $x = \frac{4}{3}$
- B. $x = -1$ and $x = -\frac{4}{3}$
- C. $x = 1$ and $x = \frac{4}{3}$
- D. $x = 1$ and $x = -\frac{4}{3}$

2 The graph of $y = f(x)$ is shown.



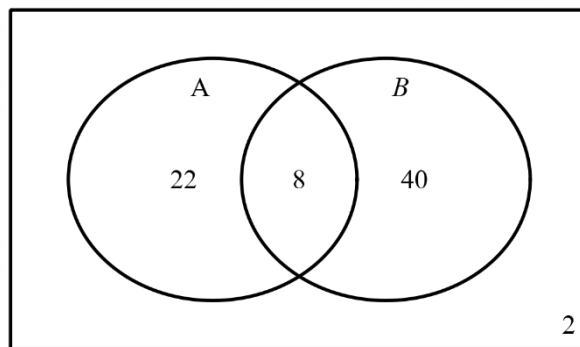
Which pair of inequalities accurately describes the sign of $f(x)$ and $f'(x)$ at the point P ?

- A. $f(x) > 0$ and $f'(x) > 0$
- B. $f(x) > 0$ and $f'(x) < 0$
- C. $f(x) < 0$ and $f'(x) > 0$
- D. $f(x) < 0$ and $f'(x) < 0$

3 How many solutions are there to the equation $\sin^2 \theta = \frac{2}{9}$ for $0^\circ \leq \theta \leq 360^\circ$?

- A. 0
- B. 1
- C. 2
- D. 4

4 Consider the Venn diagram below. An element from the sample space is selected at random.



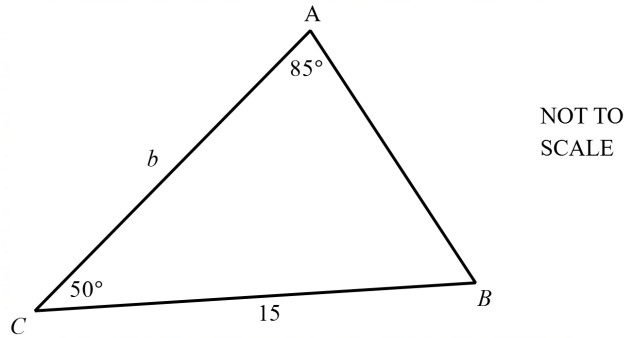
What is the value of $P(A)$?

- A. $\frac{11}{36}$
- B. $\frac{11}{35}$
- C. $\frac{5}{12}$
- D. $\frac{3}{7}$

5 Given that $g(x) = (3 - 2x)^8$, what is $g'(x)$?

- A. $g'(x) = 8(3 - 2x)^7$
- B. $g'(x) = -16(3 - 2x)^8$
- C. $g'(x) = 16(3 - 2x)^7$
- D. $g'(x) = -16(3 - 2x)^7$

- 6 Consider the triangle ABC below.



Which expression gives the value of b ?

- A. $\frac{15 \sin 45^\circ}{\sin 85^\circ}$
- B. $\frac{15 \sin 85^\circ}{\sin 50^\circ}$
- C. $\frac{15 \sin 50^\circ}{\sin 85^\circ}$
- D. $\frac{15 \sin 85^\circ}{\sin 45^\circ}$
- 7 A bag contains 10 red, 20 yellow and x green lollies. Demi chooses a green lolly and eats it.
- If she selects a second lolly at random, what is the probability that it is also green?
- A. $\frac{x}{29 + x}$
- B. $\frac{x - 1}{29 + x}$
- C. $\frac{x - 1}{30 + x}$
- D. $\frac{x}{30 + x}$
- 8 What is the gradient of the tangent drawn to the graph of $y = x^2 + 5x$ at the point $(-1, -4)$?
- A. -6
- B. -4
- C. 3
- D. 7

9 Which expression has the same value as $\sin\left(\frac{4\pi}{3}\right)$?

A. $\cos\left(\frac{\pi}{3}\right)$

B. $-\cos\left(\frac{\pi}{3}\right)$

C. $\sin\left(\frac{\pi}{3}\right)$

D. $-\sin\left(\frac{\pi}{3}\right)$

10 For what values of r does the graph of $x^2 + 4x + y^2 - 2y + 5 = r^2$ intersect the x -axis but not the y -axis?

A. $r < 1$

B. $1 < r < 2$

C. $2 < r < \sqrt{5}$

D. $r > \sqrt{5}$

Section II

60 marks

Attempt Questions 11-31

Allow about 1 hour and 45 minutes for this section.

Answer each question in the space provided. Extra writing space, if required, is located at the end of this question paper.

In Questions 11 – 31, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (2 marks)

Factorise fully $4x - xy^2$.

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Question 12 (2 marks)

Solve $|3x - 2| = 5$.

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Question 13 (1 mark)

State the domain of $y = \sqrt{x-3}$.

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Question 14 (2 marks)

Consider the functions $g(x) = 2x - 1$ and $h(x) = x^2 + 3$. Find $g(h(x))$.

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Question 15 (1 mark)

Differentiate $y = \frac{2}{x}$.

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Question 16 (1 mark)

If $\cos\theta = \sin(2\theta + 15^\circ)$, find θ for $0^\circ < \theta < 90^\circ$.

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Question 17 (2 marks)

Differentiate $f(x) = x^2 - 3x + 1$ from first principles using the formula

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$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

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Question 18 (4 marks)

A particle is moving in a straight line. Its displacement, x m, from a fixed point, O , after t seconds is given by

$$x = 2t^2 - 5t,$$

where $t \geq 0$.

- (a) Find an expression for the velocity in terms of t . **1**

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- (b) Find when the particle is at rest. **1**

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- (c) Find the velocity of the particle after 1 second. **1**

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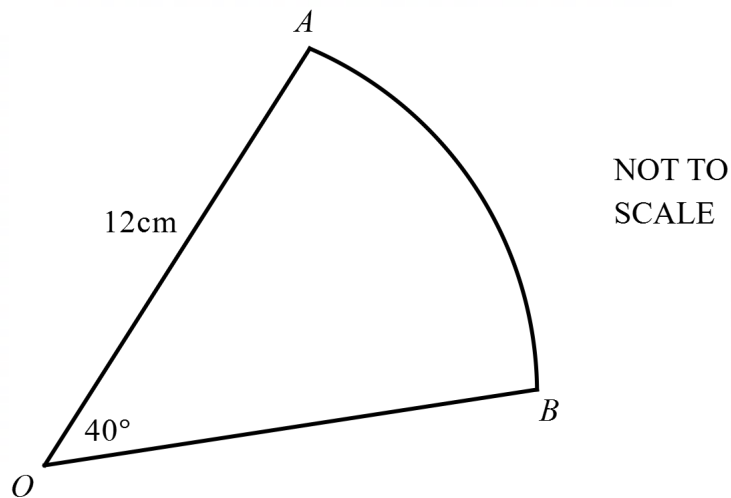
- (d) Show that the particle is moving with constant acceleration. **1**

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Question 19 (5 marks)

The sector below has centre O and a radius 12 cm. The arc AB subtends an angle of 40° at O .



- (a) Express the size of $\angle AOB$ in radians.

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- (b) Find the exact perimeter of the sector.

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- (c) Find the exact area of the sector.

2

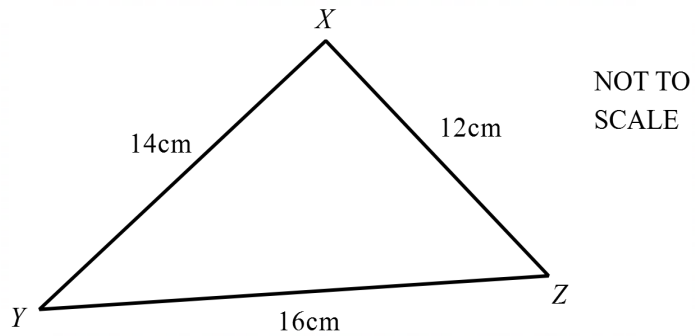
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Question 20 (2 marks)

In $\triangle XYZ$, $XY = 14$ cm, $XZ = 12$ cm and $YZ = 16$ cm.



- (a) Show that $\angle Y \approx 47^\circ$.

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- (b) Find the area of $\triangle XYZ$, to the nearest cm^2 .

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Question 21 (2 marks)

Consider the function $f(x) = 2x^2 - 3x + 2$. Show that $f(x) > 0$ for all x .

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Question 22 (3 marks)

The solutions to the quadratic equation $x^2 - 4x + 1 = 0$ are $a \pm \sqrt{b}$. Find the value of a and b . **3**

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Question 23 (2 marks)

Show that the exact value of $\sec\left(\frac{3\pi}{4}\right) = -\sqrt{2}$. **2**

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Question 24 (2 marks)

Find $\frac{d}{dx}\left(\frac{x^2}{x+1}\right)$.

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Question 25 (2 marks)

Differentiate $y = x(2-x)^5$. Give your answer in fully factorised form.

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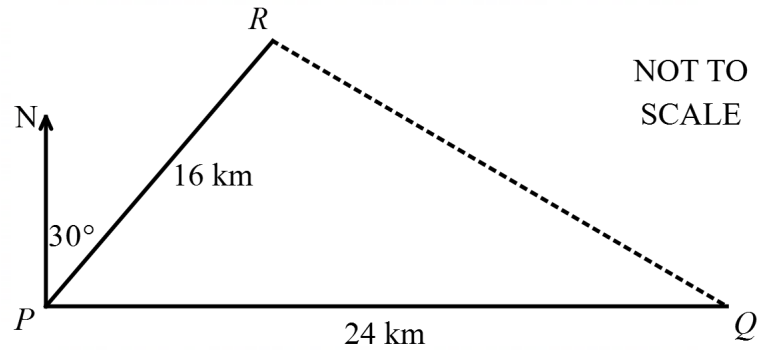
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Question 26 (5 marks)

The diagram shows Q is 24 km due east of P .

Point R is 16 km from P on a bearing of 030° .



Find the bearing of R from Q , correct to the nearest degree.

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Question 27 (3 marks)

One hundred residents were surveyed to find which types of public transport they used in the last week.

- 37 residents did not use either a train or a bus during the last week
- 43 residents used a train at least once last week
- 58 residents used a bus at least once last week

- (a) A resident is chosen at random. Using a Venn diagram, or otherwise, find the probability that the resident used a train and a bus last week.

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- (b) A resident is chosen at random. What is the probability that they used a train or a bus last week, but not both?

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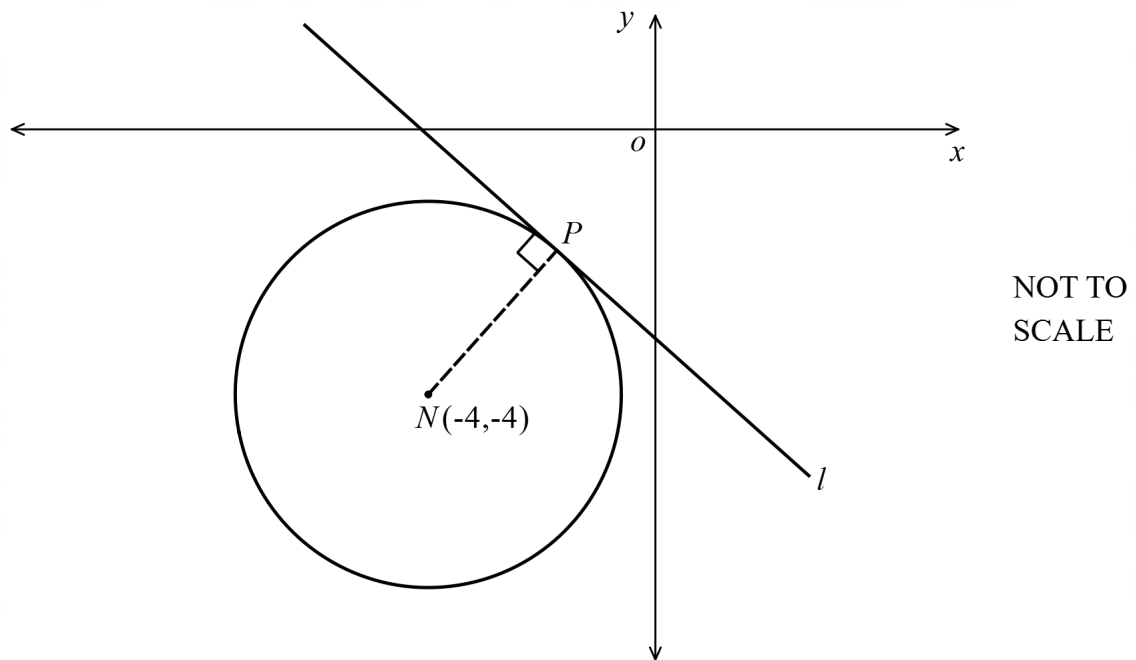
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Question 28 (6 marks)

The circle in the diagram below has centre $N(-4, -4)$.



The line l is a tangent to the circle at P and has the equation $3x + 2y + 7 = 0$.

- (a) Show that the equation of the line NP is $2x - 3y - 4 = 0$.

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(b) Find the coordinates of point P .

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(c) Find the exact radius of the circle.

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(d) State the equation of the circle.

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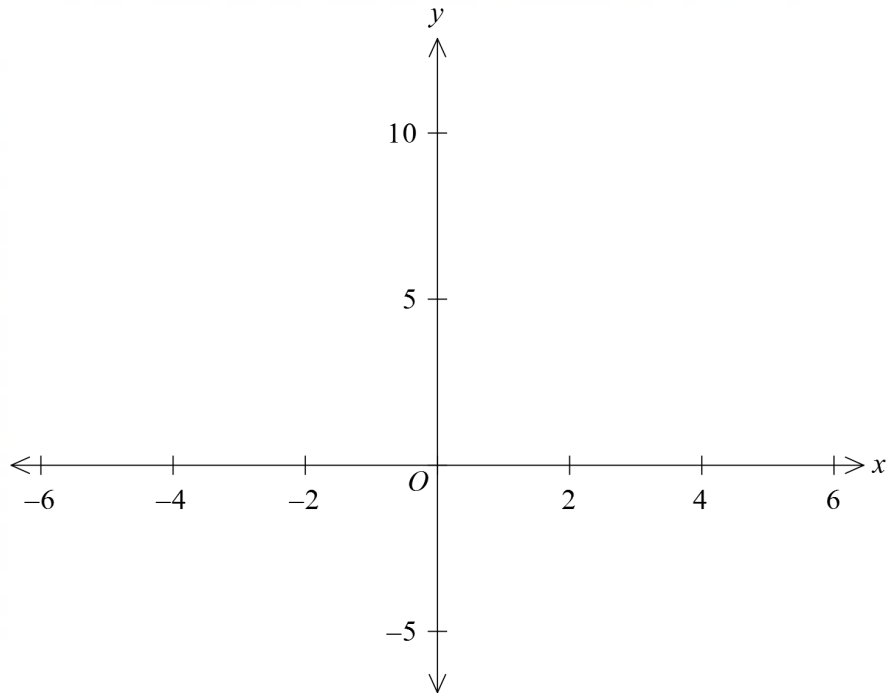
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Question 29 (4 marks)

$$\text{Let } f(x) = \begin{cases} -1, & \text{for } x < 0 \\ x^2, & \text{for } x \geq 0 \end{cases}$$

- (a) Sketch on the axes below the graph of the function $y = f(x)$.

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- (b) Is this function differentiable at $x = 0$?

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- (c) Is this function *one-to-one* or *many-to-one*? Explain your answer.

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Question 30 (4 marks)

(a) Show that $\cot^2\theta \sin\theta + \sin\theta = \operatorname{cosec}\theta$.

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(b) Hence, or otherwise, solve

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$$2\cot^2\theta \sin\theta + 2\sin\theta = -4$$

For all values of θ in the domain $[-\pi, \pi]$.

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Question 31 (5 marks)

The line l is a tangent to the curve $y = 1 - \frac{1}{x}$ at the point where $x = a$, where a is a positive constant.

If the line l passes through the origin, find the value of a **and** hence determine the equation of the line l .

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2025 11MA
Task 3



TRINITY GRAMMAR SCHOOL

Mathematics Department

S O L U T I O N S

(NESA Student Number)

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Section I

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B. $x = -1$ and $x = -\frac{4}{3}$

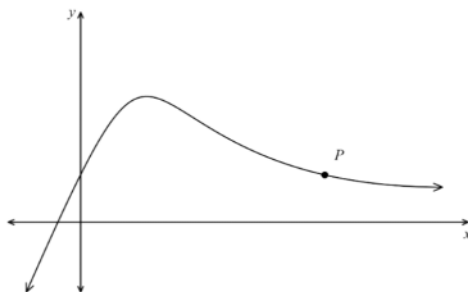
C. $x = 1$ and $x = \frac{4}{3}$

D. $x = 1$ and $x = -\frac{4}{3}$

$$x = \frac{1 \pm \sqrt{1 - 4(3)(-4)}}{2(3)}$$

$$x = \frac{1 \pm 7}{6}$$

- 2 The graph of $y = f(x)$ is shown.



Which pair of inequalities accurately describes the sign of $f(x)$ and $f'(x)$ at the point P ?

A. $f(x) > 0$ and $f'(x) > 0$

B. $f(x) > 0$ and $f'(x) < 0$

C. $f(x) < 0$ and $f'(x) > 0$

D. $f(x) < 0$ and $f'(x) < 0$

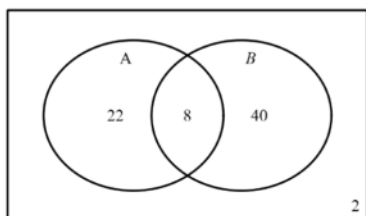
- 3 How many solutions are there to the equation $\sin^2 \theta = \frac{2}{9}$ for $0^\circ \leq \theta \leq 360^\circ$?

A. 0
B. 1
C. 2
D. 4

$$\sin \theta = \pm \frac{\sqrt{2}}{3}$$



- 4 Consider the Venn diagram below. An element from the sample space is selected at random.



What is the value of $P(A)$?

A. $\frac{11}{36}$
B. $\frac{11}{35}$
C. $\frac{5}{12}$
D. $\frac{3}{7}$

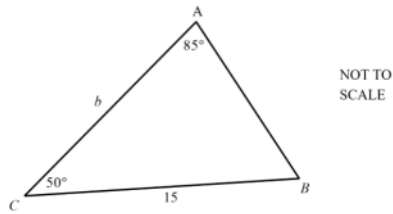
$$P(A) = \frac{30}{72} = \frac{5}{12}$$

- 5 Given that $g(x) = (3 - 2x)^8$, what is $g'(x)$?

A. $g'(x) = 8(3 - 2x)^7$
B. $g'(x) = -16(3 - 2x)^8$
C. $g'(x) = 16(3 - 2x)^7$
D. $g'(x) = -16(3 - 2x)^7$

$$g'(x) = 8(-2)(3 - 2x)^7 = -16(3 - 2x)^7$$

- 6 Consider the triangle ABC below.



Which expression gives the value of b ?

- A. $\frac{15 \sin 45^\circ}{\sin 85^\circ}$
 B. $\frac{15 \sin 85^\circ}{\sin 50^\circ}$
 C. $\frac{15 \sin 50^\circ}{\sin 85^\circ}$
 D. $\frac{15 \sin 85^\circ}{\sin 45^\circ}$

$$\frac{b}{\sin 45} = \frac{15}{\sin 85}$$

$$b = \frac{15 \sin 45}{\sin 85}$$

- 7 A bag contains 10 red, 20 yellow and x green lollies. Demi chooses a green lolly and eats it.

If she selects a second lolly at random, what is the probability that it is also green?

- A. $\frac{x}{29 + x}$
 B. $\frac{x-1}{29 + x}$
 C. $\frac{x-1}{30 + x}$
 D. $\frac{x}{30 + x}$

- 8 What is the gradient of the tangent drawn to the graph of $y = x^2 + 5x$ at the point $(-1, -4)$?

- A. -6
 B. -4
 C. 3
 D. 7

$$y' = 2x + 5$$

$$\text{at } x = -1, y' = 2(-1) + 5$$

$$= 3$$

9 Which expression has the same value as $\sin\left(\frac{4\pi}{3}\right)$?

A. $\cos\left(\frac{\pi}{3}\right)$

B. $-\cos\left(\frac{\pi}{3}\right)$

C. $\sin\left(\frac{\pi}{3}\right)$

☒ D. $-\sin\left(\frac{\pi}{3}\right)$



10 For what values of r does the graph of $x^2 + 4x + y^2 - 2y + 5 = r^2$ intersect the x -axis but not the y -axis?

A. $r < 1$

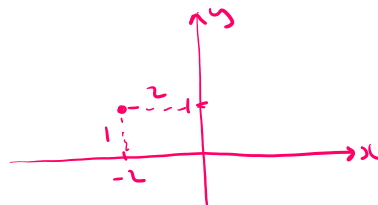
☒ B. $1 < r < 2$

C. $2 < r < \sqrt{5}$

D. $r > \sqrt{5}$

$$x^2 + 4x + 2 + y^2 - 2y + 1 + 5 = r^2 + 5$$

$$(x+2)^2 + (y-1)^2 = r^2$$



Section II

60 marks

Attempt Questions 11-31

Allow about 1 hour and 45 minutes for this section.

Answer each question in the space provided. Extra writing space, if required, is located at the end of this question paper.

In Questions 11 – 31, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (2 marks)

Factorise fully $4x - xy^2$.

2

$$= x(4 - y^2) \quad \checkmark$$

$$= x(2 - y)(2 + y) \quad \checkmark$$

Question 12 (2 marks)

Solve $|3x - 2| = 5$.

2

$$3x - 2 = 5$$

$$3x - 2 = -5$$

$$3x = 7$$

$$3x = -3$$

$$x = \frac{7}{3} \quad \checkmark$$

$$x = -1 \quad \checkmark$$

Question 13 (1 mark)

State the domain of $y = \sqrt{x-3}$.

1

$$x \geq 3 \quad \checkmark$$

Question 14 (2 marks)

Consider the functions $g(x) = 2x - 1$ and $h(x) = x^2 + 3$. Find $g(h(x))$.

2

$$\begin{aligned} g(h(x)) &= 2(x^2 + 3) - 1 \quad \checkmark \\ &= 2x^2 + 6 - 1 \\ &= 2x^2 + 5 \quad \checkmark \end{aligned}$$

Question 15 (1 mark)

Differentiate $y = \frac{2}{x}$.

1

$$\begin{aligned} y &= 2x^{-1} \\ y' &= -2x^{-2} \quad \text{or} \quad y' = -\frac{2}{x^2} \end{aligned}$$

Question 16 (1 mark)

If $\cos\theta = \sin(2\theta + 15^\circ)$, find θ for $0^\circ < \theta < 90^\circ$.

1

$$\theta + 2\theta + 15 = 90 \quad (\text{complementary } \angle\text{'s})$$

$$3\theta = 75 \quad \text{or} \quad \theta + 180 - (2\theta + 15) = 90$$

$$\theta = 25^\circ \quad \theta = 75^\circ$$

✓ 1 mark for either or both

Question 17 (2 marks)

Differentiate $f(x) = x^2 - 3x + 1$ from first principles using the formula

2

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - 3(x+h) + 1 - (x^2 - 3x + 1)}{h} \quad \checkmark$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 3x - 3h + 1 - x^2 + 3x - 1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2 - 3h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{h}(2x + h - 3)}{\cancel{h}}$$

$$= \lim_{h \rightarrow 0} 2x + h - 3$$

$$= 2x - 3 \quad \checkmark$$

Question 18 (4 marks)

A particle is moving in a straight line. Its displacement, x m, from a fixed point, O , after t seconds is given by

$$x = 2t^2 - 5t,$$

where $t \geq 0$.

- (a) Find an expression for the velocity in terms of t .

1

$$v = 4t - 5 \quad \checkmark$$

- (b) Find when the particle is at rest.

1

$$4t - 5 = 0$$

$$4t = 5 \quad t = 5/4 \text{ s} \quad \checkmark$$

- (c) Find the velocity of the particle after 1 second.

1

$$\begin{aligned} \text{when } t=1, v &= 4(1) - 5 \\ &= -1 \text{ m/s} \quad \checkmark \end{aligned}$$

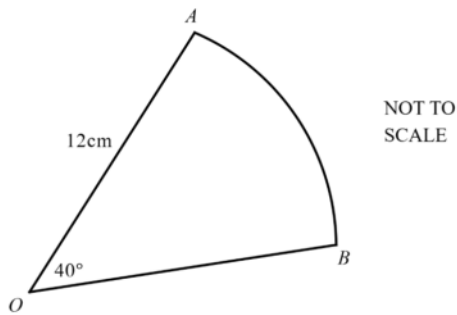
- (d) Show that the particle is moving with constant acceleration.

1

$$a = 4 \text{ m/s}^2 \quad \checkmark$$

Question 19 (5 marks)

The sector below has centre O and a radius 12 cm. The arc AB subtends an angle of 40° at O .



- (a) Express the size of $\angle AOB$ in radians.

1

$$\frac{40\pi}{180} = \frac{2\pi}{9} \quad \checkmark$$

- (b) Find the exact perimeter of the sector.

2

$$\begin{aligned} P &= 12 + 12 + 12 \left(\frac{2\pi}{9} \right) \quad \checkmark \\ &= 24 + \frac{24\pi}{9} \\ &= 24 + \frac{8\pi}{3} \text{ cm} \quad \checkmark \end{aligned}$$

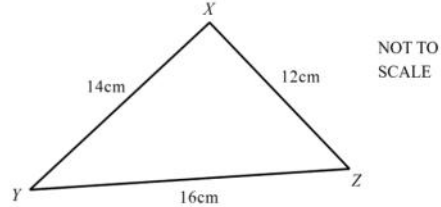
- (c) Find the exact area of the sector.

2

$$\begin{aligned} A &= \frac{1}{2} \times 12^2 \times \frac{2\pi}{9} \quad \checkmark \\ &= 16\pi \text{ cm}^2 \quad \checkmark \end{aligned}$$

Question 20 (2 marks)

In $\triangle XYZ$, $XY = 14$ cm, $XZ = 12$ cm and $YZ = 16$ cm.



- (a) Show that $\angle Y \approx 47^\circ$.

1

$$\cos Y = \frac{14^2 + 16^2 - 12^2}{2 \times 14 \times 16} \quad \checkmark$$

$$\angle Y = 46.57^\circ = 47^\circ$$

- (b) Find the area of $\triangle XYZ$, to the nearest cm^2 .

1

$$A = \frac{1}{2} \times 14 \times 16 \times \sin 47^\circ$$

$$= 82 \text{ cm}^2 \quad \checkmark$$

** check f.t for exact answer*

Question 21 (2 marks)

Consider the function $f(x) = 2x^2 - 3x + 2$. Show that $f(x) > 0$ for all x .

2

$$\Delta = (-3)^2 - 4(2)(2)$$

$$= -7 \quad \Delta < 0 \quad \checkmark$$

Concave up since $a = 2 > 0 \quad \checkmark$
 $\therefore f(x) > 0$ for all x .

positive definite
 $a > 0$
 $\Delta < 0$

Question 22 (3 marks)

The solutions to the quadratic equation $x^2 - 4x + 1 = 0$ are $a \pm \sqrt{b}$. Find the value of a and b . 3

$$\begin{aligned}x &= \frac{4 \pm \sqrt{(-4)^2 - 4(1)(1)}}{2(1)} \quad \checkmark \\&= \frac{4 \pm \sqrt{12}}{2} \\&= \frac{4 \pm 2\sqrt{3}}{2} \\&= 2 \pm \sqrt{3} \quad \checkmark \quad \checkmark \quad a=2, b=3\end{aligned}$$

Question 23 (2 marks)

Show that the exact value of $\sec\left(\frac{3\pi}{4}\right) = -\sqrt{2}$. 2

$$\begin{aligned}\sec\left(\frac{3\pi}{4}\right) &= \frac{1}{\cos\left(\frac{3\pi}{4}\right)} \quad \checkmark \\&= \frac{1}{-\cos\left(\frac{\pi}{4}\right)} \quad \checkmark \\&= \frac{1}{-\frac{1}{\sqrt{2}}} \\&= -\sqrt{2}\end{aligned}$$

Question 24 (2 marks)

Find $\frac{d}{dx} \left(\frac{x^2}{x+1} \right)$.

2

$$= \frac{(x+1)2x - x^2(1)}{(x+1)^2} \quad \checkmark$$

$$= \frac{2x^2 + 2x - x^2}{(x+1)^2} \quad \text{must simplify}$$

$$= \frac{x^2 + 2x}{(x+1)^2} \quad \checkmark = \frac{x(x+2)}{(x+1)^2}$$

$$\begin{aligned} u &= x^2 & v &= x+1 \\ u' &= 2x & v' &= 1 \end{aligned}$$

Question 25 (2 marks)

Differentiate $y = x(2-x)^5$. Give your answer in fully factorised form.

2

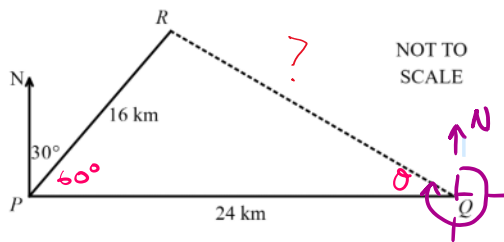
$$\begin{aligned} y' &= x \times 5(-1)(2-x)^4 + 1(2-x)^5 \quad \checkmark \\ &= -5x(2-x)^4 + (2-x)^5 \\ &= (2-x)^4 (-5x + 2-x) \\ &= (2-x)^4 (2-6x) \quad \checkmark \end{aligned}$$

$$\begin{aligned} u &= x & v &= (2-x)^5 \\ u' &= 1 & v' &= 5(2-x)^4 \times (-1) \\ & & v' &= -5(2-x)^4 \end{aligned}$$

Question 26 (5 marks)

The diagram shows Q is 24 km due east of P .

Point R is 16 km from P on a bearing of 030° .



Find the bearing of R from Q , correct to the nearest degree.

5

$$\angle RPQ = 60^\circ \checkmark$$

$$RQ^2 = 16^2 + 24^2 - 2(16)(24) \times \cos 60^\circ \checkmark$$

$$= 448$$

$$RQ = \sqrt{448} \checkmark$$

$$\cos \theta = \frac{(\sqrt{448})^2 + 24^2 - 16^2}{2 \times \sqrt{448} \times 24}$$

$$\theta = 41^\circ \checkmark$$

$$\text{Bearing} = 270 + 41$$

$$= 311^\circ \checkmark$$

Question 27 (3 marks)

One hundred residents were surveyed to find which types of public transport they used in the last week.

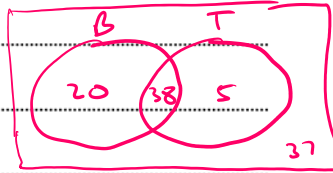
- 37 residents did not use either a train or a bus during the last week
- 43 residents used a train at least once last week
- 58 residents used a bus at least once last week

- (a) A resident is chosen at random. Using a Venn diagram, or otherwise, find the probability that the resident used a train and a bus last week.

2

$$P(T \cup B) = \frac{100 - 37}{100}$$

$$= \frac{63}{100} \checkmark$$



$$P(T \cup B) = P(T) + P(B) - P(T \cap B)$$

$$\frac{63}{100} = \frac{43}{100} + \frac{58}{100} - P(T \cap B)$$

$$P(T \cap B) = \frac{38}{100} \text{ or } 38\% \checkmark$$

- (b) A resident is chosen at random. What is the probability that they used a train or a bus last week, but not both?

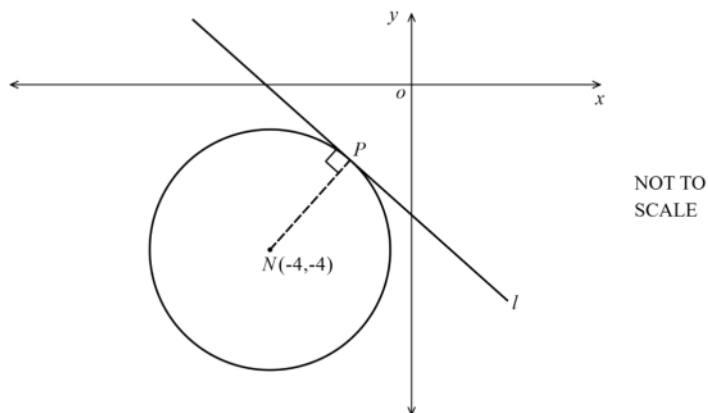
1

$$\frac{20 + 5}{100} = \frac{25}{100}$$

$$= \frac{1}{4} \text{ or } 25\% \checkmark$$

Question 28 (6 marks)

The circle in the diagram below has centre $N(-4, -4)$.



The line l is a tangent to the circle at P and has the equation $3x + 2y + 7 = 0$.

- (a) Show that the equation of the line NP is $2x - 3y - 4 = 0$.

2

$$2y = -3x - 7$$

$$y = -\frac{3}{2}x - \frac{7}{2}$$

$$m_l = -\frac{3}{2}$$

$$\therefore m_{NP} = \frac{2}{3} \checkmark$$

$$y + 4 = \frac{2}{3}(x + 4) \checkmark$$

$$3y + 12 = 2x + 8$$

$$2x - 3y - 4 = 0 \text{ as required.}$$

- (b) Find the coordinates of point P .

2

$$\begin{aligned} 2x - 3y - 4 &= 0 \quad (1) \times 3 \\ 3x + 2y + 7 &= 0 \quad (2) \times 2 \\ 6x - 9y - 12 &= 0 \quad (3) \\ 6x + 4y + 14 &= 0 \quad (4) \end{aligned} \quad \left. \begin{array}{l} (1) \times 3 \\ (2) \times 2 \end{array} \right\} \text{correct attempt } \checkmark$$
$$(4) - (3) : 13y + 26 = 0$$
$$y = -2$$
$$y = -2 \text{ into } (1) : 2x - 3(-2) - 4 = 0$$
$$2x = -2$$
$$x = -1$$
$$\therefore P(-1, -2) \checkmark \text{ must be coordinate}$$

- (c) Find the exact radius of the circle.

1

$$\begin{aligned} d &= \sqrt{(-4 - (-2))^2 + (-4 - (-1))^2} \\ &= \sqrt{4 + 9} \\ &= \sqrt{13} \quad \checkmark \end{aligned}$$

- (d) State the equation of the circle.

1

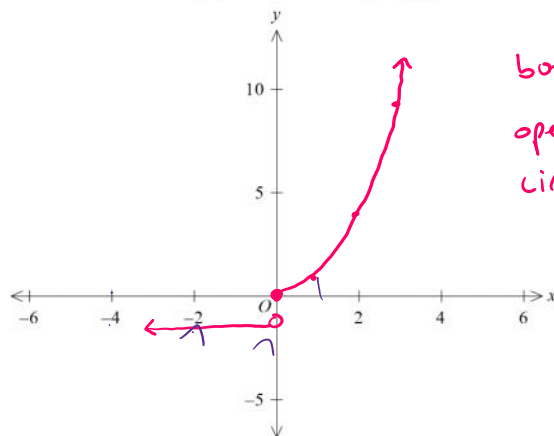
$$(x + 4)^2 + (y + 4)^2 = 13 \quad \checkmark$$

Question 29 (4 marks)

$$\text{Let } f(x) = \begin{cases} -1, & \text{for } x < 0 \\ x^2, & \text{for } x \geq 0 \end{cases}$$

- (a) Sketch on the axes below the graph of the function $y = f(x)$.

2



- (b) Is this function differentiable at $x = 0$?

1

No. ✓

- (c) Is this function *one-to-one* or *many-to-one*? Explain your answer.

1

many to one. $f(x) = -1$ maps all x values for $x < 0$ to the same y value ($y = -1$). ✓

Question 30 (4 marks)

(a) Show that $\cot^2 \theta \sin \theta + \sin \theta = \operatorname{cosec} \theta$.

2

or

$$\begin{aligned} \text{LHS} &= \frac{\cos^2 \theta}{\sin^2 \theta} \sin \theta + \sin \theta & \text{LHS} &= \sin \theta (\cot^2 \theta + 1) \quad \checkmark \\ &= \frac{\cos^2 \theta}{\sin \theta} + \frac{\sin^2 \theta}{\sin \theta} & &= \frac{1}{\operatorname{cosec} \theta} \times \operatorname{cosec}^2 \theta \quad \checkmark \\ &= \frac{1}{\sin \theta} & &= \operatorname{cosec} \theta \\ &= \operatorname{cosec} \theta & &= \text{RHS} \\ &= \text{RHS} \end{aligned}$$

(b) Hence, or otherwise, solve

2

$$2\cot^2 \theta \sin \theta + 2\sin \theta = -4$$

For all values of θ in the domain $[-\pi, \pi]$.

$$2\operatorname{cosec} \theta = -4 \quad \checkmark$$

$$\operatorname{cosec} \theta = -2 \quad \checkmark \checkmark$$

$$\frac{1}{\sin \theta} = -2$$

$$\sin \theta = -\frac{1}{2}$$

$$\angle = \frac{\pi}{6}$$

$$\begin{aligned} \text{3rd: } &\pi + \frac{\pi}{6} - 2\pi \\ &= -\frac{5\pi}{6} \end{aligned}$$

$$\theta = -\frac{\pi}{6}, -\frac{5\pi}{6} \quad \checkmark$$

$$\begin{aligned} \text{4th: } &2\pi - \frac{\pi}{6} - 2\pi \\ &= -\frac{\pi}{6} \end{aligned}$$

Question 31 (5 marks)

The line l is a tangent to the curve $y = 1 - \frac{1}{x}$ at the point where $x = a$, where a is a positive constant.

If the line l passes through the origin, find the value of a and hence determine the equation of the line l .

5

$$y = 1 - x^{-1}$$

$$y' = \frac{1}{x^2}$$

✓ 1 mark for one of these seen

$$\text{when } x = a, y' = \frac{1}{a^2} \text{ and } y = 1 - \frac{1}{a}$$

$$y - (1 - \frac{1}{a}) = \frac{1}{a^2} (x - a) \quad \checkmark$$

$$\text{at } (0, 0): 0 - (1 - \frac{1}{a}) = \frac{1}{a^2} (0 - a) \quad \checkmark$$

$$-a^2 (1 - \frac{1}{a}) = -a$$

$$-a^2 + a = -a$$

$$a^2 - 2a = 0$$

$$a(a - 2) = 0$$

$$a = 0 \text{ or } a = 2 \quad \checkmark \quad a \neq 0 \text{ since curve not defined at } x = 0$$

$$\text{when } a = 2,$$

$$y - (1 - \frac{1}{2}) = \frac{1}{4} (x - 2)$$

$$y = \frac{x}{4} - \frac{1}{2} + \frac{1}{2} \quad y = \frac{x}{4} \quad \checkmark$$

$$\text{or } x - 4y = 0$$

End of Paper

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